

Differential Splicing with Subisoform Covariance

Overview

Questions and unit of analysis

The analysis asks two distinct questions:

1. Does local splice-path usage differ by experimental condition within a cell type?
2. Does local splice-path usage differ across cell types within the same mice?

Cells are first aggregated into *mouse-by-condition-by-cell-type pseudobulks*. Biological replication is therefore at the mouse level, not the cell level.

A splice block is a local region between constitutive exonic anchors. Transcripts that take the same path through that region are collapsed to one subisoform. This targets local splicing choices, which can be identifiable even when the corresponding full-length transcripts are not.

For condition effects, each hypothesis is indexed by a splice block and a cell type. It is an omnibus test of

$$H_0 : \text{mean path usage is equal across the included conditions.}$$

Only conditions represented by at least three mouse pseudobulks are included. With two conditions and two paths, this is a precision-weighted differential-PSI test. With more conditions or paths, all condition coefficients and path log-ratios are tested jointly.

For cell-type effects, each hypothesis is indexed only by a splice block. It is an omnibus test of equal path usage across the included cell types, after absorbing mouse and condition into a mouse-specific fixed effect. Cell types must be represented in at least three mice. These two hypothesis families are reported separately.

Workflow

The analysis proceeds in five steps:

1. Construct local splice blocks from the transcript annotation and collapse transcripts to paths within each block.
2. Aggregate paired oligo(dT) and random-hexamer EC counts, and hierarchical transcript estimates, within each mouse, condition, and cell type.
3. Refine path usage in each pseudobulk with the exact paired-primer EC likelihood.
4. Estimate path-logratio covariance from EC ambiguity and UMI sampling. Estimates with weak information or boundary path usage are retained in output but excluded from Gaussian testing.

5. Fit mouse-replicate GLS models with separate inferential and between-mouse variance components. Test condition within cell type, and test cell type within mouse, as separate analyses.

Two inferential covariance calculations are reported. The *conditional* calculation fixes the fitted transcript mixture within each path. The *profile* calculation allows all transcript directions to vary as nuisance parameters. They are sensitivity analyses for the same block-by-cell-type hypothesis, not independent biological hypotheses.

Statistical Model

Paired-primer EC likelihood

For pseudobulk i and primer p , let $\mathbf{c}_{ip}$ be the EC counts, $N_{ip} = \mathbf{1}'\mathbf{c}_{ip}$, and A_p the fixed weighted EC design. For transcript abundance $\boldsymbol{\theta}_i$,

$$\mathbf{q}_{ip} = A_p \boldsymbol{\theta}_i, \quad Z_{ip} = \mathbf{1}'\mathbf{q}_{ip},$$

and the likelihood conditional on the primer-specific UMI total is

$$\mathbf{c}_{ip} \mid N_{ip}, \boldsymbol{\theta}_i \sim \text{Multinomial} \left(N_{ip}, \frac{\mathbf{q}_{ip}}{Z_{ip}} \right).$$

Oligo(dT) and random-hexamer counts have separate designs and likelihood terms, preserving their different positional and transcript-exposure profiles.

The genome-wide hierarchical model supplies a pseudobulk baseline $\boldsymbol{\theta}_i^{(0)}$. For block b , a local parameter $\boldsymbol{\delta}_{bi}$ changes the mass of all transcripts carrying path $s(t)$:

$$\theta_{it}(\boldsymbol{\delta}_{bi}) \propto \theta_{it}^{(0)} \exp\{Q'_{b,s(t)} \boldsymbol{\delta}_{bi}\}.$$

Here Q_b is a Helmert basis for the $S_b - 1$ dimensional sum-zero subspace. The total spliced mass and transcript mixture within each path are fixed during this local refinement.

Path response and inferential covariance

Let C_b map transcripts to paths. Path proportions and the response used by the differential test are

$$\psi_{bi} = C_b \frac{\boldsymbol{\theta}_{i,\text{spliced}}}{\mathbf{1}'\boldsymbol{\theta}_{i,\text{spliced}}}, \quad \mathbf{y}_{bi} = Q'_b \log \psi_{bi}.$$

For two paths, y_{bi} is the log path-odds divided by $\sqrt{2}$. Under the local parameterization, $\mathbf{y}_{bi} = \mathbf{y}_{bi}^{(0)} + \boldsymbol{\delta}_{bi}$.

The conditional covariance is the inverse Fisher information for $\boldsymbol{\delta}_{bi}$. For the profile calculation, write $\boldsymbol{\eta}_i = \log \boldsymbol{\theta}_i$. The expected transcript information is

$$I_{\eta,ip} = N_{ip} \left[\frac{\text{diag}(\boldsymbol{\theta}_i) A'_p \text{diag}(\mathbf{q}_{ip}^{-1}) A_p \text{diag}(\boldsymbol{\theta}_i)}{Z_{ip}} - \mathbf{r}_{0,ip} \mathbf{r}'_{0,ip} \right],$$

where $\mathbf{d}_p = A'_p \mathbf{1}$ and $\mathbf{r}_{0,ip} = (\mathbf{d}_p \odot \boldsymbol{\theta}_i) / Z_{ip}$. Information is added across primers. If $J_{bi} = \partial \mathbf{y}_{bi} / \partial \boldsymbol{\eta}'_i$, then

$$V_{bi}^{\text{profile}} = J_{bi} \left(\sum_p I_{\eta,ip} \right)^+ J'_{bi}.$$

The implementation checks whether J_{bi} overlaps the information null space. A non-estimable contrast is reported as unidentified rather than being assigned zero variance by a pseudoinverse.

Mouse-replicate differential tests

Within one cell type and block, let i index mouse pseudobulks and let X_i encode condition:

$$\hat{\mathbf{y}}_{bi} = X_i B_b + \boldsymbol{\epsilon}_{bi}, \quad \text{Var}(\boldsymbol{\epsilon}_{bi}) = V_{bi} + \tau_b^2 I.$$

The EC-derived V_{bi} measures inferential uncertainty. The scalar τ_b^2 measures residual between-mouse variation and is estimated by REML under the relevant null model. GLS uses $(V_{bi} + \tau_b^2 I)^{-1}$.

For the *condition test*, one model is fit for each block and cell type. X_i contains an intercept and condition indicators, and the Wald statistic tests all non-reference condition coefficients. Condition labels are permuted with their group sizes fixed for calibration.

For the *cell-type test*, one model is fit for each block across cell types. X_i contains mouse fixed effects and cell-type indicators. Because condition is constant within a mouse, the mouse effects also absorb condition and all other shared mouse-level shifts. The Wald statistic tests all cell-type indicators. Calibration permutations shuffle cell-type labels within each mouse, preserving the pairing and observed missingness pattern.

The primary p-value is the asymptotic Wald value because 20 permutations do not resolve the tail required for genome-wide FDR control. Benjamini–Hochberg correction is applied separately for each combination of test axis (condition or cell type) and covariance method (conditional or profile).

Implementation Details

Splice-block construction

GTF exons are represented by zero-based, half-open intervals. Exon boundaries define atomic intervals; intervals exonic in every annotated spliced transcript are merged into constitutive anchors. Consecutive anchors define a block, and a transcript path is its ordered exon-interval chain between them. Terminal pseudo-anchors include alternative first and last regions. A gene without a shared constitutive region receives one whole-gene block.

Blocks with one path are discarded. Simple two-path blocks include skipped exons, mutually exclusive exons, alternative donors and acceptors, and retained introns. The implementation also supports nested and multi-path blocks. Precursor features remain nuisance features in the EC model but are not assigned to spliced paths.

Pseudobulks and local fitting

Cell transcript proportions from the hierarchical decoder are weighted by cell UMI totals and aggregated by mouse, condition, and cell type. Pseudobulks must contain at least 20 cells and 10^5 total UMIs. A block is fit when its gene has at least 25 gene-assigned UMIs in at least two qualifying pseudobulks. Only ECs whose compatible transcripts map uniquely to that gene enter the local likelihood.

Bounded L-BFGS optimizes the paired multinomial likelihood, with $|\delta| \leq 20$ to prevent floating-point underflow. Two-path fits are initialized from a log-ratio grid because the paired-primer normalizers can make the objective non-concave.

Identifiability and reliability

Finite Fisher covariance is not sufficient near a simplex boundary. Parametric bootstrap identified two reliability requirements:

$$\lambda_{\max}(V_{bi}) \leq 1/8 \quad \text{and} \quad \min_s \hat{\psi}_{bis} \geq 0.15.$$

Thus every tested log-ratio direction has effective information of at least eight and every fitted path has at least 15% usage. All estimates are written, together with convergence, identifiability, and reliability flags; only reliable estimates enter GLS.

Results

Condition-test scope

The primary analysis used the 25-gene-UMI cutoff and blocks with at most eight represented paths. It fit 10,100 blocks and 359,713 pseudobulk-by-block path estimates; 359,711 fits converged. After reliability filtering, there were 829 conditional and 506 profile tests.

These 1,335 rows represent 829 distinct block-by-cell-type hypotheses, covering 527 distinct splice blocks in eight cell types. The profile analysis is available for a subset of the conditional hypotheses. The possible conditions were 5xFAD, 5xFIRE, 5xFIRE PBS, 5xFIRE transplant, FIRE, and WT. For the 829 conditional hypotheses, 275 compared two conditions, 155 compared three, 153 compared four, 130 compared five, and 116 compared all six. The included set varies because every condition in a test must have at least three independent mouse pseudobulks for that block and cell type.

Condition-test results

No block was significant at 5% FDR in either covariance analysis. At the unadjusted $p < 0.05$ level:

- the conditional analysis identified 24 block-by-cell-type hypotheses, covering 21 distinct blocks;
- the profile analysis identified 20 block-by-cell-type hypotheses, covering 17 distinct blocks;
- 20 hypotheses were nominally significant in both analyses.

The union is 24 nominal block-by-cell-type hypotheses covering 21 blocks, but the corrected number of significant hypotheses and blocks is zero. Thus there is no FDR-significant evidence for condition-dependent splicing within cell type. The two covariance variants should not be added as if they were separate discoveries.

Cell-type effects within mouse

The paired cell-type analysis tested 369 blocks with conditional covariance; 218 of these also had a valid profile-covariance test. Across blocks, 12 cell types were represented, with each individual block comparing between two and 11 cell types. Every included cell type had at least three mice, and the mouse-fixed-effect design was required to have full rank.

Cell-type differences were readily detected:

- 57 conditional tests were nominally significant and 22 remained significant at 5% FDR;
- 28 profile tests were nominally significant and five remained significant at 5% FDR;

- all five profile discoveries were among the 22 conditional discoveries.

The union therefore contains **22 FDR-significant splice blocks** with evidence of different path usage across cell types. Permuting cell-type labels within mouse gave conservative null rejection rates of 2.0% for conditional and 1.4% for profile covariance.

Calibration

Across the 25-UMI analysis, 2.7% of permuted-label Wald p-values were below 0.05 for both conditional and profile covariance, indicating conservative null calibration. A parametric bootstrap of 3,000 reliable local fits gave mean squared standardized error 1.10 and 94.5% coverage for nominal 95% intervals. Independent Gaussian simulations of REML-GLS gave type-I error between 2.7% and 5.0% across the tested biological-variance settings.

Numerical tests additionally verify skipped-exon block construction, identifiable covariance after collapsing indistinguishable transcripts, rejection of information-null path contrasts, the analytic binomial Fisher variance, and recovery of imposed GLS effects.

Coverage-cutoff sensitivity

UMIs	blocks	tests	conditional null $p < .05$	profile null $p < .05$	bootstrap coverage
25	10100	1335	0.027	0.027	0.945
50	5896	755	0.031	0.030	0.936
100	2820	283	0.029	0.025	0.946
200	1034	69	0.028	0.026	0.948
500	195	10	0.030	0.050	0.946

No cutoff produced an FDR-significant block. The 500-UMI calibration estimate is based on only 100 permutation tests.

For reliable two-condition, two-path conditional tests, mean asymptotic power for a 0.5 ILR effect was 0.49, 0.51, 0.44, 0.56, and 0.59 at increasing cutoffs. The final two values are selected from only eight and two eligible tests. Among 103 hypotheses shared by the 25- and 50-UMI analyses, power was 0.453 and 0.449. Lower coverage cutoffs therefore increase the number of testable blocks without degrading calibrated precision among shared tests. The 25-UMI cutoff is the default.

Limitations

The profile covariance is conditional on the fitted genome-wide decoder. Uncertainty in shared decoder parameters is expected to be small for common paths but may matter for rare genes or cell types. A mouse-level jackknife of the hierarchical fit is the preferred extension.

The current biological variance is isotropic in log-ratio space. Estimating a full covariance separately for every block would be unstable with few mice. A later empirical-Bayes model could shrink path-specific biological covariances across genes.